PARISHRAM 2.0

CLASS 12TH 2026

Sexual Reproduction in Flowering Plants

Biology

Botany | LECTURE-7

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TOPICS TO BE COVERED

1. Pollen Pistil Interaction
2. Double Fertilization
3.
4.

Pollen Pistil Interaction



Carpel / pistil → ♀ sex organ

→ As soon as pollengrain lands on stigma, pistil begins the **compatibility test** i.e whether the pollengrain is of right-type [of same species] or not through exchange of chemical dialogues

Hence it is a dyanamic process

∴ Pollination doesnot guarantee fertilization

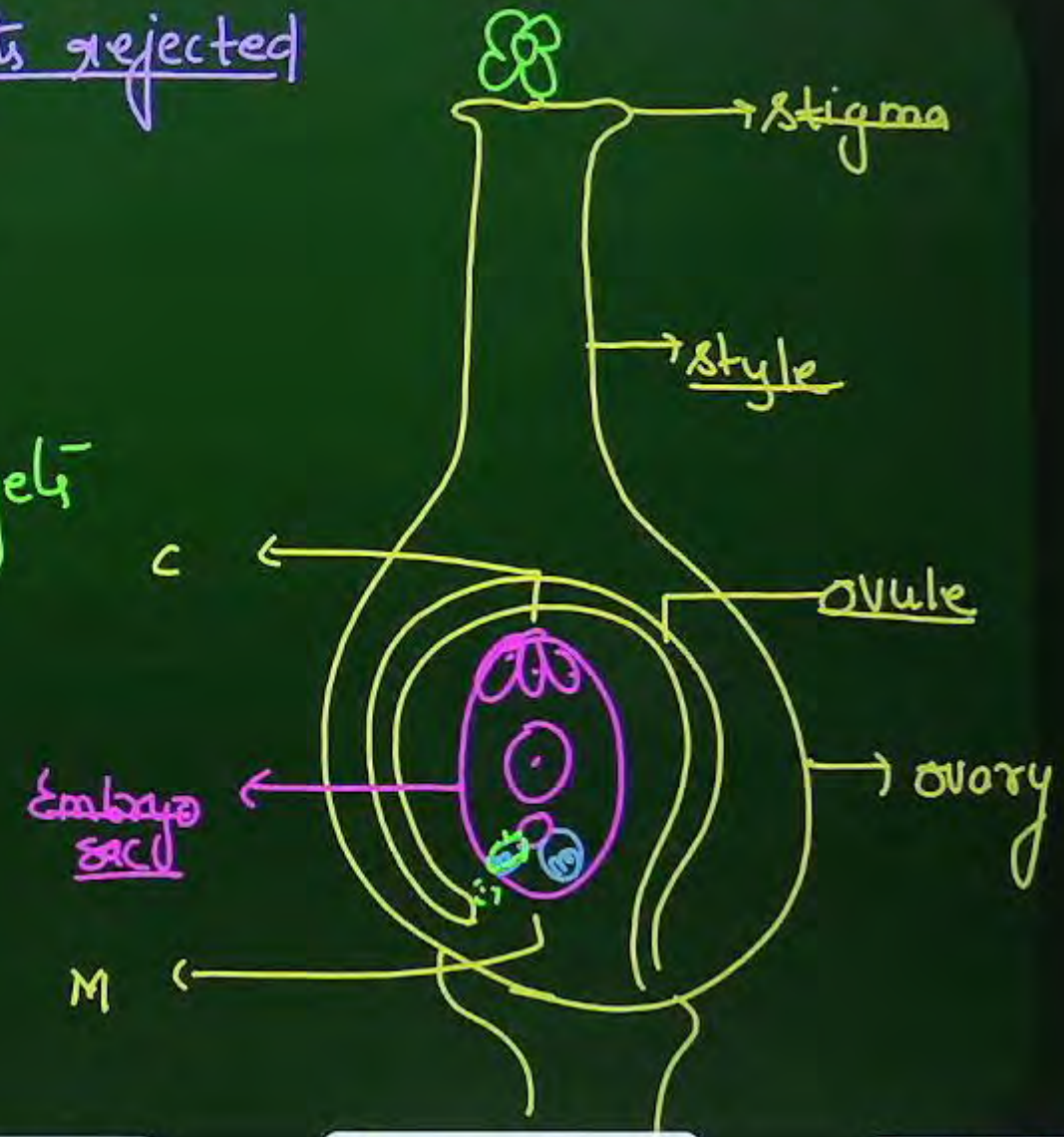
↳ If suppose PG is of wrong-type, then it will fail the test & hence will not germinate i,e it gets rejected

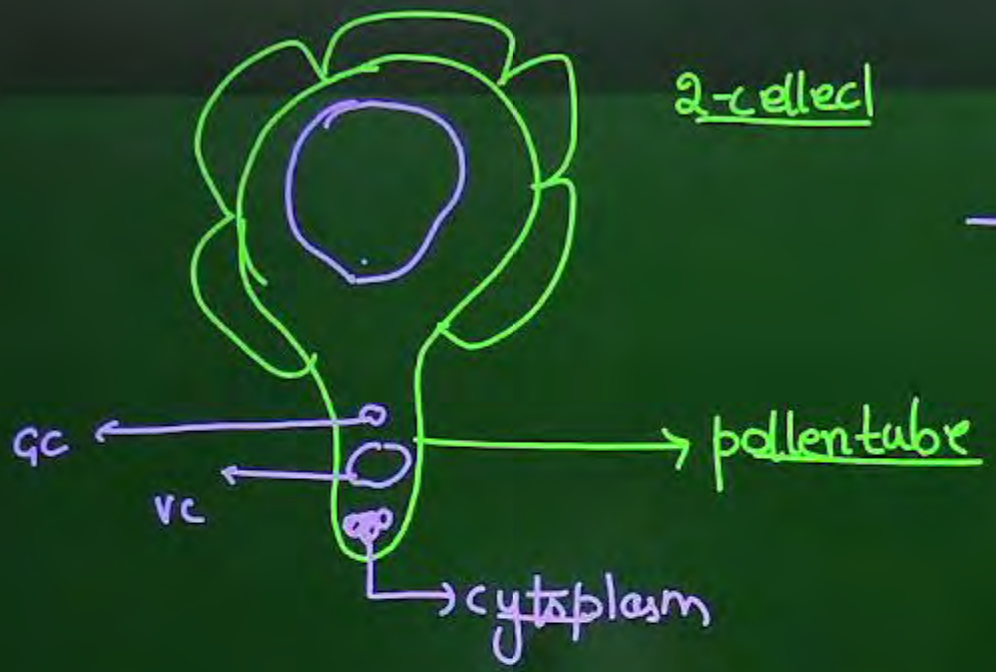
However if (PG) passes the test.

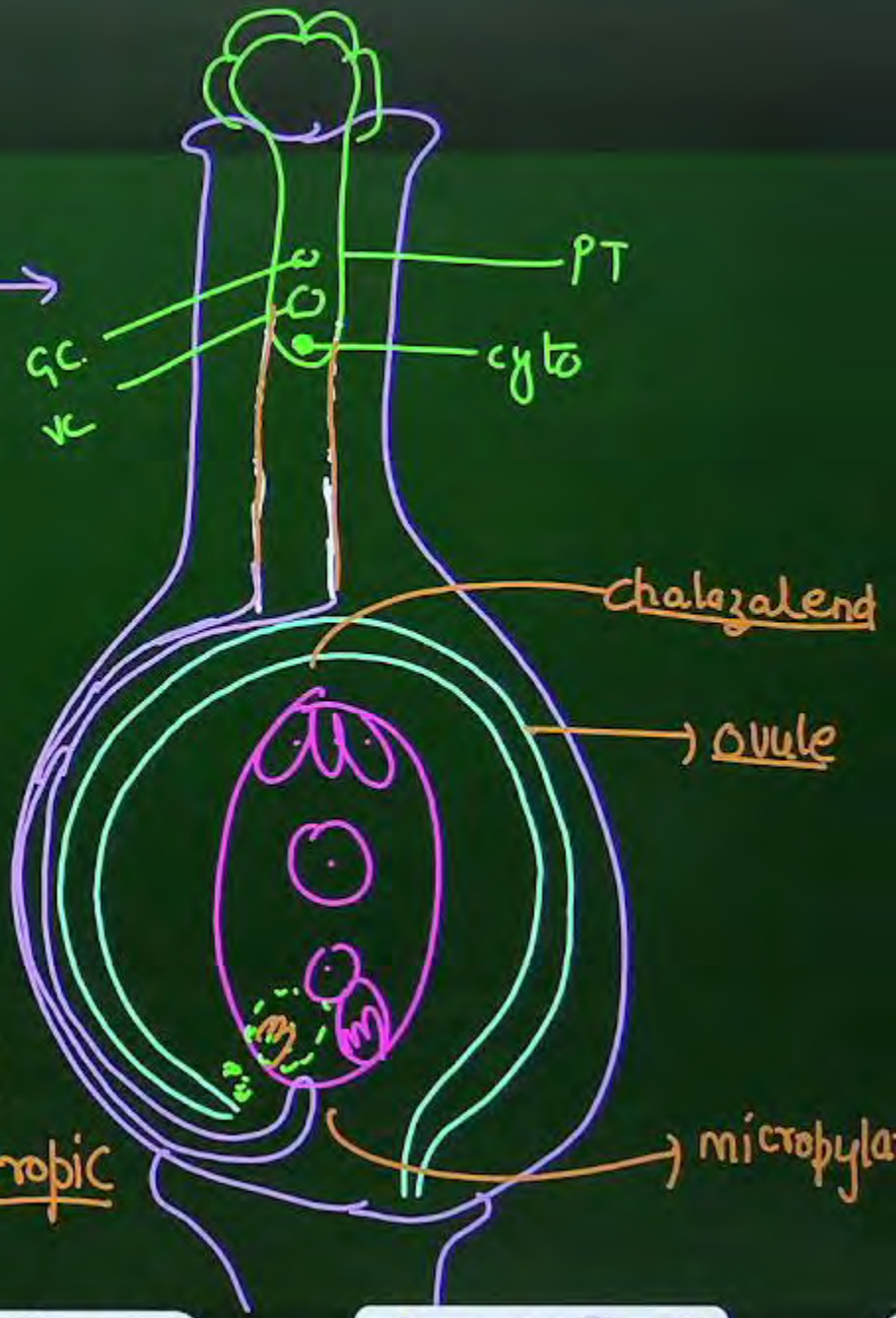
↓

Out of 2 synergid, one gets degenerated to release chemicals

↳ Once Pollengrain passes the test, its intine bulges out through germpore to form pollentube



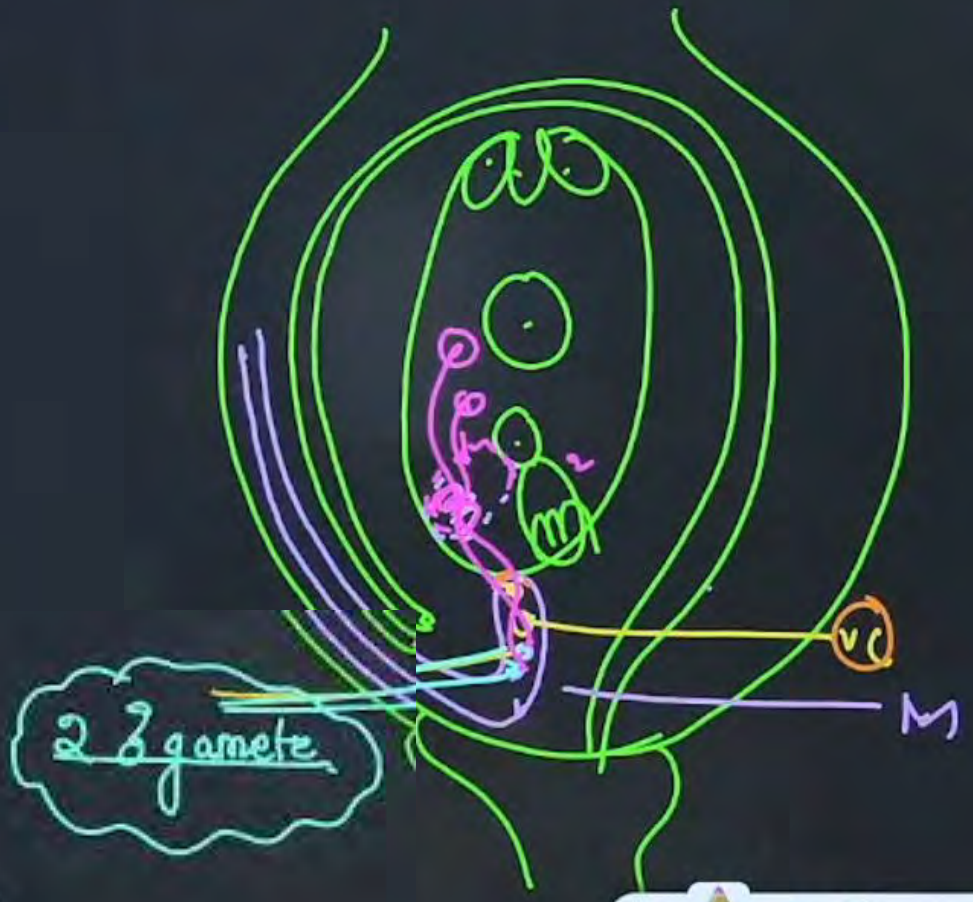




Note:- Pollen tube grows in the style towards ovary under the influence of chemical secreted by degenerated synergid → chemotropic

→ Pollen tube enters the ovule through micropyle under the influence of chemical (porogamy)

Till here it is pollen-pistil interaction



→ After Pollen tube enters the ovule, its tip it touches the embryosac, thus due to friction tip rupture. Now cytoplasm + VC degenerates.

Pollen Pistil Interaction



Carpel / pistil → ♀ sex organ

→ As soon as pollen grain lands on stigma, pistil begins the **compatibility test** i.e whether the pollen grain is of right type [of same species] or not through exchange of chemical dialogues

Hence it is a dynamic process

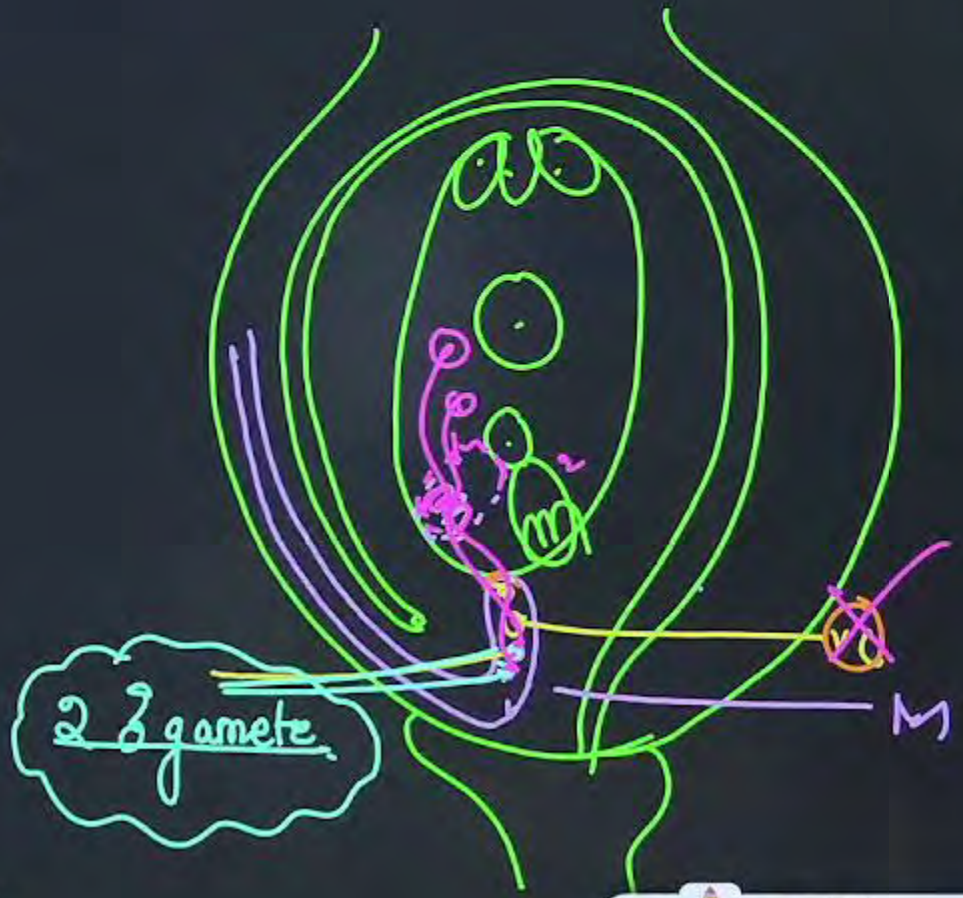
∴ Pollination doesnot guarantee fertilization

→ Pollen tube enters the ovule through micropyle under the influence of chemical

(porogamy)

Till here it is pollen-pistil interaction

⟹



→ After Pollen tube enters the ovule, its tip it touches the embryosac, thus due to friction tip rupture. Now cytoplasm + VC degenerates.

18:02



→ Large 2n central cell

→ ♀ gamete

Double fertilization

1st ♂ + ♀ → syngamy → Zygote

(n) (n)

diploid cell

+

2nd ♂ / n + 2n central cell → Triple fusion → Primary endosperm cell (3n)

by Nawaschin in Fritillaria

QUESTION

Where is the proximal end of the filament attached?

P. To the ovary

Q. To the petal or thalamus

R. To the stigma

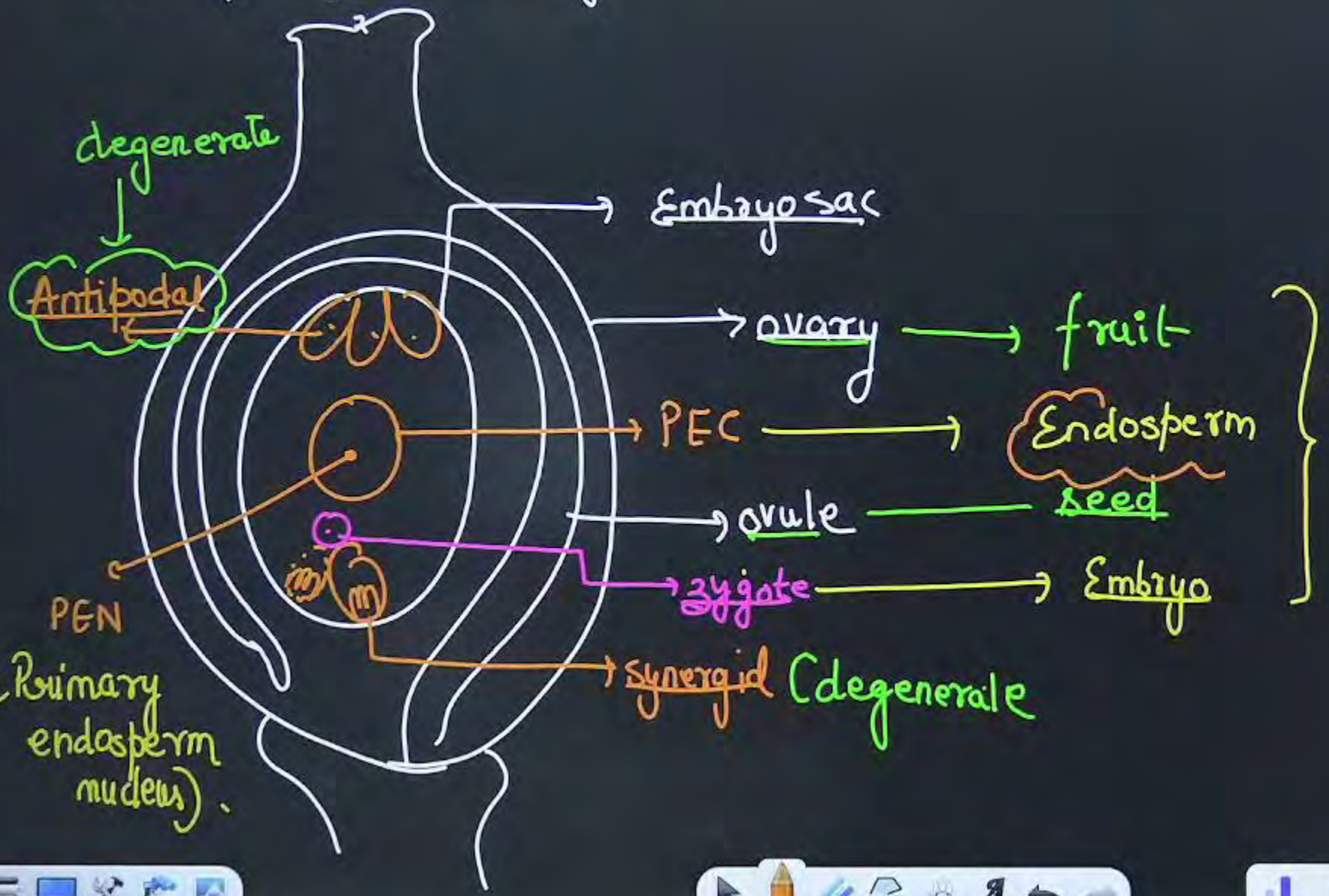
S. To the sepal



1. P only
2. P and Q
3. Q only
4. P, R and S

Post-fertilization changes

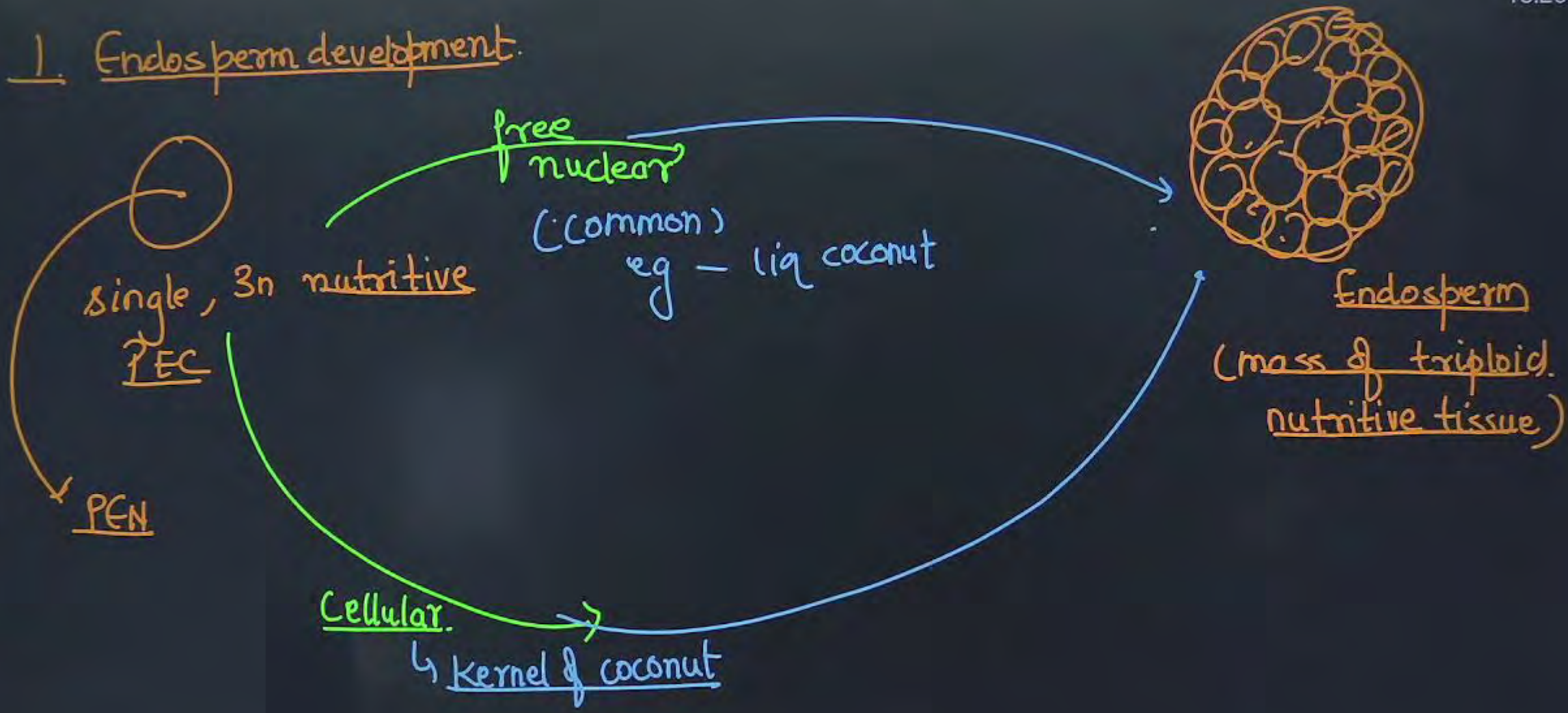
PEC
(Primary endosperm cell)



= Post fertilization changes

18:26

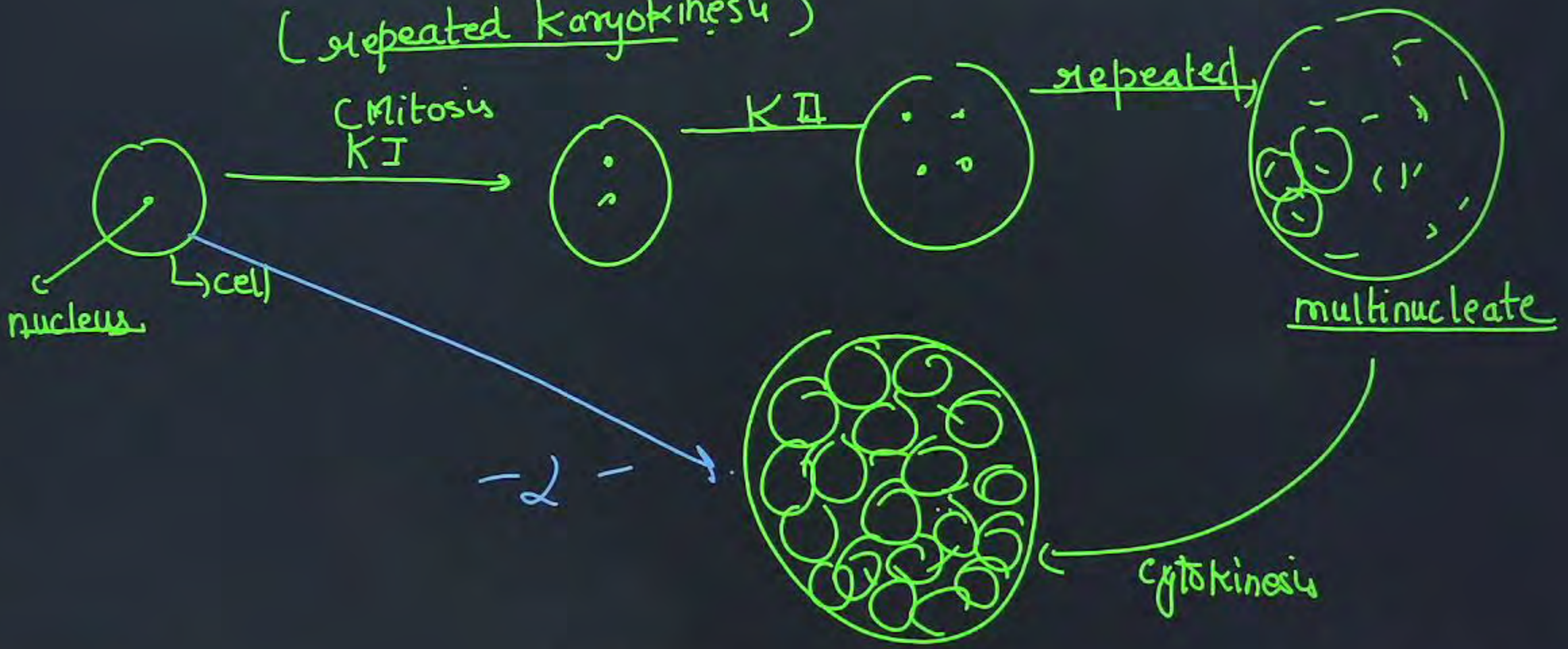
1. Endosperm development



18:29

Free nuclear mitosis.

(repeated karyokinesis)



THANK YOU